Case 03 · Research and POC: Removal Sample Audit

Following the Magshimim "Research and POC" template. The POC was written in Python and actually run; the numbers are its output. What is real and what is simulated is stated explicitly.

Basic research: sequence of events

- **Draw:** from a public seed announced in advance, 1,500 deletions are drawn from a district's removal list.
- **Assign:** each case goes to a field checker.
- **Check:** the checker visits the address and marks either "lives here and is eligible" (a wrong deletion) or "deletion correct" (dead, shifted, duplicate).
- **Estimate:** share of wrong deletions with a 95% interval, overall and per reason.
- **Decide:** districts above a threshold (e.g. 5%) get a full re-check.

Basic research: components

Sampler (reproducible draw), Field App (offline), Estimator (intervals), Dashboard (district map).

Basic research: security challenges

- **Biased draw:** defence: public seed, published method.
- **A checker who does not check:** marks "correct" without visiting. The central risk; lab 4 measures it.
- **A checker with an interest:** if checkers belong to the machinery that deleted, they have a reason to confirm.

Basic research: comparison with the leading alternative (booth-officer house visits)

Today's visits are not random and produce no summary number or margin of error; ours are random from a public seed, give one rate with a range, and check 1,500 cases per district rather than hundreds of thousands.

Basic research: ethical issue

A figure like "11% wrong" can become a political headline, so the range, method and seed are always published with it.

Five key features

- **Draw from a public seed:** `random.Random(seed).sample`.
- **Stratification by reason:** allocation by stratum size.
- **Wilson estimate:** rate and 95% interval from k of n.
- **Coverage test:** on a synthetic district with known truth, how often the interval contains it.
- **Checker sensitivity:** probability that a checker "confirms" a wrong deletion, and the resulting bias.

The POC: research question

Does a sample of a few hundred deletions per district give an accurate, reliable estimate of the wrong-deletion rate, and how sensitive is it to who checks in the field?

The POC: technologies and limits

Pure Python (`random`, `math`). **Important limit:** districts, deletions and true error rates are synthetic (a district of 120,000 deletions; 70% in the "absent, shifted or other" bucket; true error rates: dead 3%, duplicate 8%, bucket 19%). The statistics are real.

The POC: results

Lab	Measured	Result
Precision	sample size vs margin (truth 14.8%)	100: ±8.4 · 600: ±2.8 · 1,500: ±1.7 · 3,000: ±1.3
Coverage	2,000 audits of 1,500 cases	interval contained the truth 94.8% of the time (95% expected), mean ±1.8
Stratify	one reason vs three	dead 3.2% (truth 3.0), duplicate 9.2% (8.3), bucket 19.4% (18.9)
Field	checker "confirms" part of the wrong deletions	0%: 14.8% (95% coverage) · 10%: 13.2% (63%) · 25%: 11.0% (1.8%) · 40%: 8.7% (0%)

The POC: conclusions

- ✅ **Precision and coverage:** proven; 1,500 cases give about ±1.8 points, and the interval is honest.
- ✅ **Stratification:** proven, and it shows the opaque bucket is where most errors sit.
- ⚠️ **Field:** the finding that follows the idea from here. The estimate is only as good as the checker. If a quarter of wrong deletions are "confirmed" without a real check, the estimate drops from 14.8% to 11% and the interval contains the truth only 1.8% of the time. The statistics do not protect against this; they only look safe.

Conclusion: ✅ survives into specification. Open question: who is the ground truth in the field.